Moment maps & Hamiltonian reduction

§1 Poisson algebras/mfds.

Def. A Poisson algebra is a commutative algebra A with a bi-derivation $\{\cdot,\cdot\}: A^{\otimes 2} \to A$ s.t. $(A, \{\cdot,\cdot\})$ is a Lie algebra.

Def. $I \trianglelefteq A$ is a Poisson ideal if $\{I, A\} \subseteq I$.

- A/I is a Poisson algebra.
- Exercise Guess the definition of homomorphism of Poisson algebras.
- Assume G is a group and A is a G-rep. and $\forall g \in G$, $\rho_A(g)$ is hom-m. Then A^G is a Poisson subalgebra. (Exercise)
- We will think of algebras of functions.
- Let $Y \subset X$, $\mathcal{F}(X)$ - Poisson. (X - Poisson mfd)
  Y is Poisson submfd if $\mathbb{I}(Y)$ is a Poisson ideal
  in that case $\mathcal{F}(Y) = \mathcal{F}(X)/\mathbb{I}(Y)$ - Poisson algebra $\Rightarrow Y$ - Poisson mfd.
- Let $G \curvearrowright X$ ~~$\not\curvearrowright$~~ by Poisson maps. $\Rightarrow G \curvearrowright \mathcal{F}(X)$ then
  " $\mathcal{F}(X/G) = \mathcal{F}(X)^G$ - Poisson algebra $\Rightarrow$ " X/G - Poisson mfd. "
- Poisson structure induces the map $\varphi: \mathcal{F}(X) \to \mathrm{Vect}(X)$, $f \mapsto \{f, \cdot\}$.

Exercise: Check that it is a hom-m.

- Vector fields in the image of φ are called Hamiltonian.

§2 Symplectic mfds.

Def. A symplectic mfd. is a smfd. X with a two-form $\omega \in \Lambda^2 X$ s.t.

- ω is non-deg. at every point.
- $d\omega = 0$.

- ω induces isom-s $T_x X \longrightarrow T_x^* X$ $\forall x \in X$ through $v \longmapsto \omega(v, \cdot)$. Denote this Ψ_x.
- Define $\forall f, g \in \mathcal{F}(X)$, $\{f, g\}(x) := df(\Psi_x^{-1}(dg)) = \omega(\Psi_x^{-1}(df), \Psi_x^{-1}(dg))$
- Denote $\Psi_x^{-1}(df) =: v_f$.

Exercise $\{\cdot,\cdot\}$ is a Poisson bracket. Moreover, $\Psi_x^{-1} \circ d_x = \{ev_x \circ \varphi$.

Example Consider any mfd/vty M. Denote $X = T^*M$.

We have $\pi: X \longrightarrow M$ - natural projection.

Define $\theta \in T^*X$ by $\theta_\alpha(v) = \alpha(d_\alpha \pi(v))$ (Here $\alpha \in T^*_{\pi(\alpha)} X$)

In local coordinates:

$(x_1, \dots, x_n, p_1, \dots, p_n) \Rightarrow \pi: (x_1, \dots, x_n, p_1, \dots, p_n) \longmapsto (x_1, \dots, x_n)$

($x_1, \dots, x_n$: coords on M; $p_1, \dots, p_n$: linear coords in fiber)

$\alpha = \sum_{i=1}^n p_i dx_i$

If $v = \sum \xi_i \frac{\partial}{\partial x_i} + \sum \eta_i \frac{\partial}{\partial p_i}$, $d_\alpha \pi(v) = \sum_i \xi_i \frac{\partial}{\partial x_i} \Rightarrow$

$\Rightarrow \theta_\alpha(v) = \left(\sum_{i=1}^n p_i dx_i\right)\left(\sum_i \xi_i \frac{\partial}{\partial x_i}\right) = \sum_{i=1}^n p_i \xi_i \Rightarrow$

$\Rightarrow \theta = \sum_{i=1}^n p_i dx_i$.

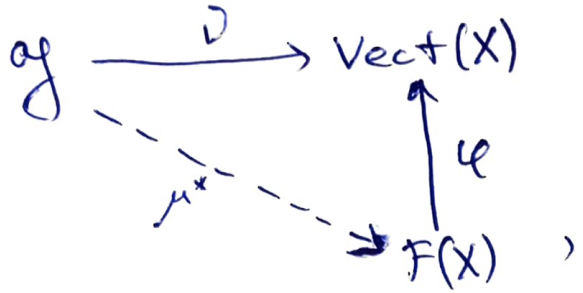
Set $\omega = d\theta$.

Exercise: ω is a symplectic form on X.

§3. Group actions and moment maps.

Def.

Let G be a connected, simply-connected Lie group (algebraic).
Assume $G \curvearrowright X$ - Poisson mfd. by Poisson transformations.
The action is called Hamiltonian if there exists a lift. (homomorphism)



where $\nu : \xi \longmapsto \left(x \mapsto \frac{d}{dt}\left(e^{t\xi} \cdot x\right)\Big|_{t=0}\right)$

$\varphi : f \longmapsto \{f, \cdot\}$

Prop. Hamiltonian action is Poisson.

Proof (sketch)

It is sufficient to check $\forall \xi \in \mathfrak{g}$, $\mathcal{L}_{\nu(\xi)}(\Pi) = 0$.

$$\mathcal{L}_{\nu(\xi)}(\Pi) = \sum_{i,j} \mathcal{L}_{\nu(\xi)}\left(\{x_i, x_j\}\partial_i \wedge \partial_j\right) = \sum_{i,j} \Big(\mathcal{L}_{\nu(\xi)}\left(\{x_i, x_j\}\right)\partial_i \wedge \partial_j +$$

$$+ \{x_i, x_j\}\left[\nu(\xi), \partial_i\right] \wedge \partial_j + \{x_i, x_j\}\partial_i \wedge \left[\nu(\xi), \partial_j\right]\Big) = 0$$

$$\nu(\xi) = \sum_k \nu(\xi)_k \partial_k = -\{H_\xi, x_k\}\partial_k$$

$$\mathcal{L}_{\nu(\xi)}\left(\{x_i, x_j\}\right) = -\{H_\xi, \{x_i, x_j\}\}$$

Def.

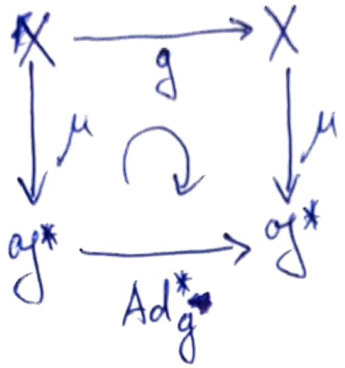
Let $G \curvearrowright X$ be a Hamiltonian action.

The moment map is $\mu : X \longrightarrow \mathfrak{g}^*$

$$x \longmapsto (\xi \longmapsto (\mu^*(\xi))(x))$$

Prop. Let $G \curvearrowright X$ be a Hamiltonian action.

$\forall x \in M, \forall g \in G$



Proof.

$\forall \xi, \eta \in \mathfrak{g} \quad \frac{d}{dt}\Big|_{t=0} \left(\langle \mu(e^{t\xi} \cdot x), \eta \rangle \right) = \{ \langle \mu, \xi \rangle, \langle \mu, \eta \rangle \}(x) =$

$= \langle \mu(x), [\xi, \eta] \rangle = \langle -ad^*_\xi(\mu(x)), \eta \rangle$

$\Rightarrow \frac{d}{dt}\Big|_{t=0} \left(\mu(e^{t\xi} \cdot x) \right) = -ad^*_\xi(\mu(x)) \Rightarrow$

$\Rightarrow \frac{d}{dt} \left(\mu(e^{t\xi} \cdot x) \right) = -ad^*_\xi \left(\mu(e^{t\xi} \cdot x) \right)$

$\frac{d}{dt} \left(Ad^*_{e^{t\xi}}(\mu(x)) \right) = -ad^*_\xi \left(Ad^*_{e^{t\xi}}(\mu(x)) \right)$ ← Same ODE with initial condition $\mu(x)$.

$\Rightarrow \mu(e^{t\xi}, x) = Ad^*_{e^{t\xi}}(\mu(x))$ $\Rightarrow$ Holds $\forall g \in G$.

∎

Exercise 1.

Given a symplectic mfd X with the 2-form ω show that

$\{f,g\} := \omega(v_f, v_g)$ defines a Poisson structure.

Here v_f is s.t. $\omega(v_f, \cdot) = df(\cdot)$.

Exercise 2.

Show that the two-form $\sum_{i=1}^{n} dp_i \wedge dx_i$ on the vector space

$\langle \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}, \frac{\partial}{\partial p_1}, \dots, \frac{\partial}{\partial p_n} \rangle =: V$ is non-degenerate.

Exercise 3.

Let $\{.,.\}$ be a Poisson bracket. Let $(x_1, \dots, x_n)$ be local coordinates on X. Show that for any $f, g \in C^\infty(M)$,

$$\{f,g\} = \sum_{i,j=1}^{n} \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial x_j} \{x_i, x_j\} \Rightarrow \{.,.\} = \sum_{i,j} \{x_i, x_j\} \frac{\partial}{\partial x_i} \wedge \frac{\partial}{\partial x_j}.$$

Exercise 4

Given that $\mathcal{V}(\xi)$ (vector field) acts on functions by $\mathcal{V}(\xi)(f) = \{f, \mu^*(\xi)\}$.

(a) Compute coordinates of $\mathcal{V}(\xi)$ w.r.t. local chart $(x_1, \dots, x_n)$.

(b) Given that

$\mathcal{L}_{\mathcal{V}(\xi)}(f) = \mathcal{V}(\xi) \quad \forall f \in C^\infty(X)$,

$\mathcal{L}_{\mathcal{V}(\xi)}(V) = [\mathcal{V}(\xi), V] \quad \forall V \in \mathrm{Vect}(X)$

and Leibnitz rule for $\mathcal{L}_{\mathcal{V}(\xi)}$

$$\mathcal{L}_{\mathcal{V}(\xi)}\left(\sum_{i,j} \{x_i, x_j\} \frac{\partial}{\partial x_i} \wedge \frac{\partial}{\partial x_j}\right) = 0.$$

show

Exercise 5

Let $G = GL_n$, $\mathfrak{g} = \mathfrak{gl}_n$. Coadjoint action is defined by

$\mathrm{Ad}^*_g(\lambda)(\xi) = \langle \lambda, \mathrm{Ad}_{g^{-1}}(\xi) \rangle \quad \forall \lambda \in \mathfrak{g}^*, \xi \in \mathfrak{g}, g \in G$

$\langle \mathrm{ad}^*_\xi(\lambda), \eta \rangle = -\langle \lambda, [\xi, \eta] \rangle \quad \eta \in \mathfrak{g}$

Show that

a) $\frac{d}{dt}\big|_{t=0} \left(\mathrm{Ad}^*_{e^{t\xi}}(\lambda)\right) = \mathrm{ad}^*_\xi(\lambda)$

b) $\frac{d}{dt}\left(\mathrm{Ad}^*_{e^{t\xi}}(\lambda)\right) = \mathrm{ad}^*_\xi\left(\mathrm{Ad}^*_{e^{t\xi}}(\lambda)\right)$.